Lane-Changing Strategy for Autonomous Vehicle With Adaptive Adjustment of Decision-Making Preference Based on Game Theory

Liu Yang, Chengdeng Cao, Qianxi Zhao, Junzhang Yang, and Ailong Fan

***Abstract*—Lane-changing is a critical task in autonomous driving. Rational lane-changing behavior can enhance both travel efficiency, driving safety. However, existing studies suffer from issues such as ambiguous causal relationships in decision-making logic and insufficient generalization capabilities in complex scenarios. To address these challenges, this paper proposes an interactive modeling approach that combines game theory with deep reinforcement learning (DRL). This method integrates the scene generalization ability of DRL and the clear decision-making causal relationship of game theory into a unified model. Furthermore, to tackle the challenge of determining optimal lane-changing timing, this study employs a long short-term memory (LSTM) neural network to enable game theory to incorporate temporal considerations. In this model, the agent intervenes in the calculation of the game theory-based payoff function and adjusts decision-making preferences based on indicators such as efficiency, safety, and comfort to align with environmental requirements, thereby adapting to diverse environments. Tests demonstrate that the model proposed in this paper is capable of achieving safe and efficient decision-making by adaptively adjusting decision-making preferences in complex and extreme scenarios. Specifically, it achieves a success rate of 99.67% in an environment with surrounding vehicles exhibiting diverse driving styles. Overall, the model presented in this paper not only ensures efficient and safe decision-making through preference adjustments but also preserves the clear causal relationships inherent in game-theoretic decision-making.**

***Index Terms*—Lane-changing strategy, autonomous vehicle, decision-making preference, game theory, deep reinforcement learning (DRL).**

I. Introduction

FUNCTIONAL autonomous driving includes three functions: environmental perception [1] and recognition [2], vehicle control [3], and decision-making [4]. Among them, autonomous driving decision-making is the "brain" of an autonomous vehicle (AV). Improper decision-making may to lead to traffic congestion or even result in traffic accident [5]. Consequently, decision-making algorithms for AVs have emerged as a central focus in recent years within autonomous driving research. These algorithms commonly address critical issues such as lane-changing, merging, and collision avoidance [4], with lane-changing frequently selected as a standard scenario for AV testing [6]. However, the current lane-changing decision-making algorithms still face numerous challenges, including a single driving style, difficulty in adapting to complex and dynamic environments, unclear decision-making causal relationships, and difficulty in handling interactions among different traffic participants. Therefore, it is crucial to improve the generalization ability of decision-making and ensure clear causal relationships in the decision-making process in complex lane-changing scenarios while simultaneously achieving adaptability of driving styles to the environment.

Due to the significance of lane-changing decisions, many researchers have conducted studies on the lane-changing decision of AVs. At the beginning, many researchers used rule-based approaches to describe lane-changing decision-making [7], [8]. Although using rule-based approaches to describe lane-changing behavior can achieve certain results, it is challenging to extract lane-changing rules in a complex traffic environment [9], making it difficult to adapt to such an environment. Moreover, describing lane-changing decision-making using rule-based approaches overlooks the interaction between vehicles. Furthermore, numerous researchers have developed data-driven decision-making models [10], [11]. While leveraging extensive datasets enables these models to manage complex traffic scenarios effectively, this approach also presents several challenges, including the requirement for vast amounts of data, high computational costs, slow update cycles, and limited interpretability [12].

Many researchers have also investigated alternative approaches, including decision-making methods grounded in deep reinforcement learning (DRL) and game theory [13], [14]. In the domain of autonomous driving, DRL has demonstrated superior performance compared to the aforementioned modeling techniques [15]. DRL combines the robust comprehension capabilities of deep neural networks with the core trial-and-error learning

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mechanism of reinforcement learning. In autonomous driving decision-making, DRL agent can continuously interact with the traffic environment to fit safe and efficient decision-making [16]. This encompasses various branches such as standard DRL [17], safe reinforcement learning [18], distributional reinforcement learning [19], and hierarchical reinforcement learning [20]. These distinct branches can address certain problems from specific perspectives. Safe reinforcement learning refines the optimization criterion by integrating constraints into the objective function of existing policy optimization algorithms [21]. Distributional reinforcement learning enhances model performance by optimizing for worst-case scenarios instead of focusing solely on average performance [22]. Hierarchical reinforcement learning improves model performance through task decomposition into multiple sub-tasks, thereby reducing the search space [23]. Hole et al. [24] combined distributed reinforcement learning with ensemble methods and proposed a decision-making model capable of coping with perceptual uncertainty, which can balance risk and efficiency in occlusion scenarios. Wang et al. [25] proposed a highway lane-changing model based on DRL, mapping the state input to the outputs of acceleration and direction angle to achieve efficient vehicle control. Although self-driving cars can make diverse decisions based on the current environment and achieve excellent results (such as lane keeping, vehicle following, and lane changing), most DRL models map the input directly to the output of the decision without understanding the causal relationship in the decision-making process.

Unlike DRL, the game theory approach models the interaction among rational decision-makers. Models based on game theory mainly make decisions according to the respective payoffs of AVs and surrounding vehicles (SVs). Both the calculation of payoff and the derivation of decision-making have clear mathematical formulas. Therefore, the decision-making model based on game theory has a clear causal relationship and distinct decision-making logic [26], providing a valuable framework for studying the behaviors of multiple vehicles. Many researchers have employed game theory to analyze the interactions and decision-making processes between AVs and SVs, achieving relatively satisfactory outcomes [27], [28]. Lu et al. [29] factored in traffic trend within the gaming process and combined neural networks for intent recognition with game theory, resulting in an improved lane-changing decision-making model. Arbis et al. [30] incorporated conflict risk into payoff matrix construction to encourage cautious driving by AVs as a means of risk reduction. In game theory, diverse payoff functions can be designed so that AVs can separately consider factors such as driving safety and efficiency during decision-making processes. Assigning varying degrees of importance to these considerations enables different driving style preferences in autonomous driving decision-making (e.g., aggressive, normal, conservative) [31], [32]. While game theory-based approaches demonstrate significant advantages in addressing lane-changing decision-making including logical clarity and interaction with surrounding vehicles, previous studies relied on preset benefit functions for altering decision-making preferences without adapting to complex and dynamic environments.

Furthermore, some researchers have conducted interactive modeling to address the disadvantages of game theory and other methods, thereby enhancing the model's effectiveness. Lopez et al. [33] designed a neural network with a single hidden layer based on the payoff function and independent variables of game theory, and trained it using reinforcement learning to obtain a model that can only make lane-changing decision-making. Liu et al. [34] designed a decision-making model using a neural network based on the imitation payoff function and were able to make timely decision-making according to the different aggression levels of surrounding vehicles. Yuan et al. [35] combined the K-level game and DRL. By training a DRL model with an adaptive game level in scenarios with different game levels, it achieves high security. Xu et al. [36] combined graph games with offline reinforcement learning and used RL to derive the optimal strategy in graph games to achieve robust path tracking control. Although these methods retain the interactive nature of game theory, they lose the characteristic of having clear causal relationships in the decision-making process. In addition, Chan et al. [37] combined RL with non-cooperative games. By using RL to reduce the solution space and simplify two-player games, the system was finally able to interact with different human drivers. Although this method preserves the clear causal relationships inherent in game theory's decision-making process, autonomous driving decision-making typically requires consideration of temporal dependencies. However, it is difficult for game theory to consider the interaction trajectories of surrounding vehicles during the decision-making process [38].

To summarize, employing parameters to describe lane-changing decision-making and utilizing a large volume of data to train decision-making models both have inherent limitations. The sole reliance on DRL has problems such as unclear causal relationships in the decision-making process, while the isolated use of game theory fails to adapt to dynamic environments and does not adequately account for the temporal dependency of lane-changing. Moreover, human drivers tend to exhibit varying decision-making preferences across different environments, rather than adhering to a singular and fixed pattern. The integration of game theory and DRL for interaction modeling allows for the full utilization of their respective advantages, enabling dynamic adjustment of decision-making preferences in response to environmental changes and adapting to the dynamic and complex environment [39]. For the aforementioned problems, this paper takes into account the temporal dependency of lane-changing and interactively models game theory and DRL. This approach ensures that the decision-making process of the model possesses a clear causal relationship, the capability to interact with surrounding vehicles, and the adaptability to complex dynamic environments. Then, relevant indicators, including risk and efficiency, were incorporated to rigorously test and validate the model. The primary contributions of this paper can be summarized as follows:

1) Regarding the challenge of game theory's limited adaptability to dynamic environments, this paper proposes an interactive modeling approach that integrates DRL with the Stackelberg game. Specifically, DRL does not directly make decisions but instead intervenes in the computation

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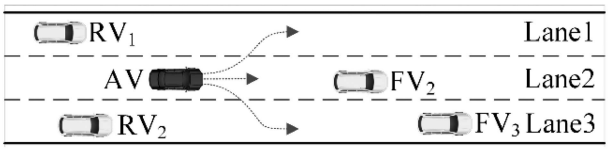

Fig. 1. Lane-changing environment.

of the payoff function by observing the surrounding environment, thereby influencing the decision-making process. Game theory, as the core framework for decision-making, preserves the clear causal relationships in the decision-making process while enhancing the model's adaptability to dynamic environments through DRL (The adaptive realization of driving style to the environment is achieved through DRL intervention based on game theory.). Moreover, to strengthen the coupling in the interactive modeling process, this paper derives the reward function of DRL from the payoff function, thereby ensuring the alignment of objectives between game theory and DRL.

2) Regarding the problem that game theory finds it difficult to capture the temporal dependency of lane-changing, this paper integrates the long short-term memory (LSTM) Network into the DRL agent. The agent evaluates the long-term interactions between the AV and surrounding vehicles, thereby determining the weights of each indicator in the payoff function during its computation. Consequently, the temporal dependency of lane-changing is explicitly accounted for in the reward function calculation, influencing the decision-making process.

The remainder of this paper is organized as follows. The following Section introduces the problem description, Section III describes the model framework and method description of lane-changing decision-making. The model of lane-changing decision-making is constructed in Section IV. The training and testing results are demonstrated in Section V. Finally, Section VI concludes this paper.

II. Problem Description

A. Lane-Changing Decision-Making of AVs

The lane-changing scenario is illustrated in Fig. 1, the decision-making process of the AV is influenced by both the Front Vehicle (FV) and the Rear Vehicle (RV). For example, if FV_2 is traveling at a slower velocity, impacting AV's movement, AV must assess whether to change lanes or maintain its current lane while adjusting its velocity accordingly. This decision-making process is affected by the driving behavior of surrounding vehicles, including RV_1, RV_2, FV_1, FV_2, and FV_3 as well as by AV's own driving style. If AV adopts an aggressive driving style, it may attempt to overtake FV_2 in order to follow with faster vehicles in the adjacent lane and wait for an opportunity to pass. Conversely, if AV adopts a conservative driving style, it may decelerate and follow behind FV_2 without changing lanes, prioritizing safety over driving efficiency.

TABLE I
Indicator Importance of Driving Styles [30], [42], [43]

Driving styles	Travel efficiency	Ride comfort	Driving safety
Aggressive	80%	10%	10%
Conservative	10%	20%	70%
Normal	30%	20%	50%

B. Driving Style

In various scenarios, distinct driving styles result in diverse driving experiences. These styles place varying levels of emphasis on driving safety, ride comfort, and travel efficiency [40], and different driving styles can be reflected in the relative proportions among driving safety, ride comfort, and travel efficiency [41]. By adjusting the proportional relationship among driving safety, ride comfort, and travel efficiency in game theory, conservative, normal, and aggressive driving styles can be achieved [30]. There are three different driving styles as follows:

1) *Aggressive:* This driving style gives top priority to travel efficiency. To achieve the driving purpose, the driver may adopt some aggressive behaviors.
2) *Conservative:* This driving style gives priority to driving safety. Generally, it involves decelerating and minimizing lane changes to follow the FV as closely as possible.
3) *Normal:* This driving style is between the aggressive style and the conservative style, aiming to strike a balance among different driving goals.

This paper constructs a game payoff function comprising three indicators: travel efficiency, ride comfort, and driving safety. These indicators are quantified using information such as velocity, acceleration, and position. By adjusting the weight coefficients of these three indicators, different priorities can be assigned to achieve varying driving styles. For instance, an aggressive driving style emphasizes higher priority on travel efficiency. According to prior works by other researchers [30], [42], [43], the weight coefficients of the three indicators for the three types of driving styles are presented in Table I. These coefficients have been validated as feasible within these works. Fang et al. [26] also obtained a similar proportion for the three indicators by using inverse reinforcement learning in the trajectory dataset. Furthermore, the GT-PPO-LSTM proposed in this paper changes the relative importance of these three indicators in real time based on the environment and historical information, thereby adapting to the complex and dynamic environment. That is to say, the AV in this paper cannot be measured by any of these three driving styles, but will adaptively adjust to the driving style most suitable for the current environment based on the environmental information.

III. Model Framework and Method Description of Lane-Changing Decision-Making

In this section, we primarily present the framework of the model and elaborate on the role of each component. Given that Proximal Policy Optimization (PPO) demonstrates stable performance [16] and the Stackelberg game more accurately

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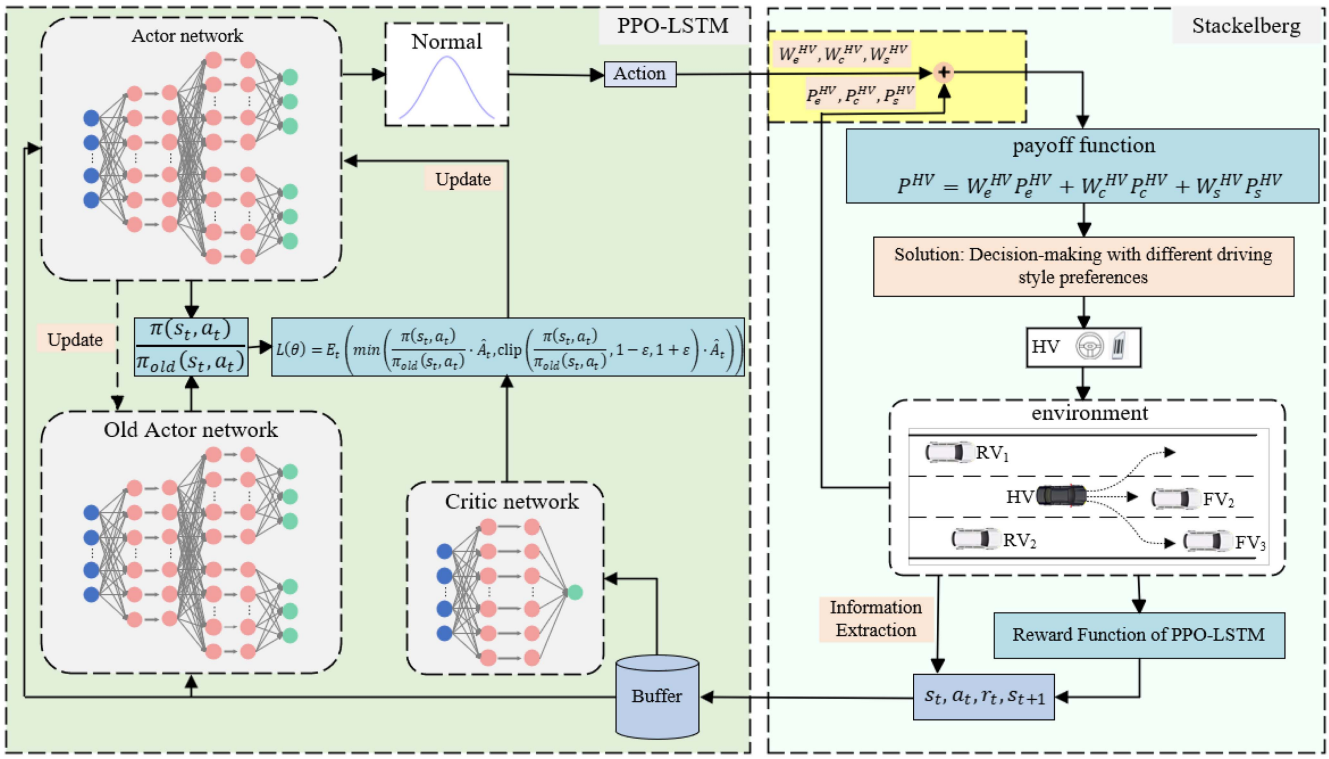

Fig. 2. Model framework of decision-making.

captures the interactions between vehicles, in this paper, these two approaches are integrated with LSTM for modeling (GT-PPO-LSTM). A detailed description of the model is provided below.

A. Model Framework of Lane-Changing Decision-Making

In the Stackelberg game, the interaction process between the AV and the SV is clear, and the causal process of decision-making is also clear. In DRL, the agent can make decisions that adapt to the surrounding environment by observing it and can also incorporate recurrent neural networks to address the temporal dependency of lane-changing. Therefore, in this paper, by combining the advantages of game theory and DRL, interaction modeling between DRL and game theory was conducted, leading to the realization of GT-PPO-LSTM. This model takes the game theory framework as its core and can consider lane-changing temporal dependency while adaptively adjusting decision-making preferences to adapt complex dynamic environments. Fig. 2 shows the framework of GT-PPO-LSTM.

In Fig. 2, the left part represents PPO-LSTM (green area). The input integrates information from SVs and AV, specifically including acceleration, velocity, and position data (as detailed in Part IV, Section C). Then, by sampling from the Gaussian distribution, three outputs were obtained: the safety term coefficient, the efficiency term coefficient, and the comfort term coefficient of the game theory payoff function. The sum of these three coefficients was equal to 1. This part primarily determines the extent to which safety, efficiency, and comfort should be considered by AV during the game process. This determination is based on the current information and historical information of both SVs and the AV itself, which corresponds to the computation of the payoff function in game theory. Additionally, this part takes into account the temporal dependency of lane changes. By incorporating these elements, the adaptability of game theory to dynamic and complex environments is significantly enhanced.

The right part of Fig. 2 represents the game theory component (blue area), and the game theory component is the main body of the decision-making in the GT-PPO-LSTM. This part takes the information of SVs and AV as inputs, calculates the driving safety payoff, travel efficiency payoff, and ride comfort payoff using (1)–(8), and subsequently determines the total payoff of AV (the yellow area on the right side of Fig. 2) by integrating the coefficients of the three payoffs in the PPO-LSTM module. Finally, the game theory solution determines whether to change lanes and the longitudinal acceleration, which are then calculated as the steering wheel angle and acceleration through quintic polynomial path planning. This section is primarily accomplished through game theory calculations. From the input of environmental information to the output of decision-making, every step is grounded in specific equations. This approach provides a transparent overview of the payoffs associated with each component and the solution calculation process, particularly when an incident occurs and requires tracing. Consequently, it clarifies the causal relationships within the decision-making process.

The left and right parts in Fig. 2 work together to achieve a model capable of making lateral and longitudinal collaborative decision-making. In the decision-making process of this model, the main framework is still based on game theory,

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which can maximize the clarity of causal relationships in the decision-making process. In this model, DRL is innovatively integrated into game theory to enhance the model's ability to adapt to complex dynamic environments. This method can retain the advantages of strong interpretability of game theory and clear causal relationships in the decision-making process to the greatest extent. This model enables AVs to adapt the importance of different indicators in diverse scenarios to fit the decision-making preferences of diverse driving styles. It dynamically adjusts the relative importance of travel efficiency, driving safety, and ride comfort based on diverse scenarios (e.g., assigning higher weight to efficiency when it becomes more crucial). Additionally, it utilizes game theory for interactions with surrounding vehicles and adapts decision-making preferences based on the current environment (e.g., adopting a more cautious strategy during heavy traffic flow), thereby achieving diverse driving style preference decision-making and holding significant practical implications.

B. Game Theory in the Model Framework of Lane-Changing Decision-Making

Game theory serves as a robust tool for studying the interactions among decision makers, wherein a game is a well-defined mathematical object comprising players, strategies, and payoffs. At present, there are many types of game theory, within which the Stackelberg game can better describe the relationship between vehicles in this paper. Therefore, the Stackelberg game is used in this paper. The Stackelberg game is also known as the master-slave game. In this game, the leader has the advantage of leadership and can occupy advantageous position, while the follower is required to make decisions after the leader. The AV can predict the reactions of surrounding vehicles and thereby decide whether it is possible to change lanes. This decision-making process can be accurately described as a Stackelberg game, in where AV assumes the role of the leader while RVs act as followers. AV makes its decision after carefully considering the response of RVs, which subsequently determines its own course of action in order to maximize individual benefits. In this paper, AV takes into account both FVs and RVs when considering lane changes, enabling it to identify opportune moments to switch lanes and enter gaps between vehicles, thereby enhancing overall lane-changing safety. Throughout the decision-making process in game theory, the logic behind decision-making is transparent and readily acceptable. It is noteworthy that a smaller payoff is considered more favorable in this paper. It is important to emphasize that the model proposed in this paper has the ability to interact with multiple vehicles.

C. DRL in the Model Framework of Lane-Changing Decision-Making

DRL, combining deep learning and reinforcement learning, enables the agent to continuously engage in trial and error, maximizing rewards to generate optimal behavioral policies. Simultaneously, employing a neural network to represent the strategy function or value function enables the agent to adapt effectively to various state and action space problems. Currently,

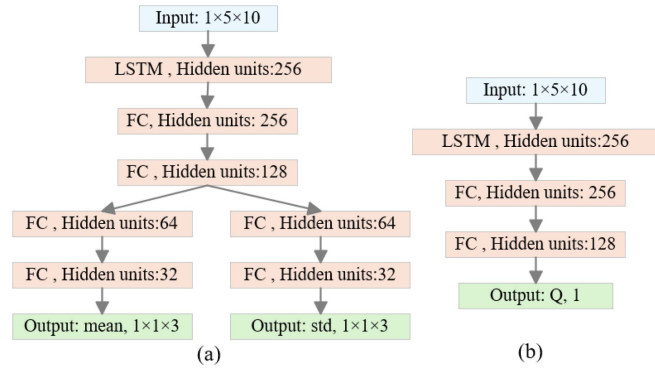

Fig. 3. Network structure. (a) Actor network. (b) Critic value network.

there exist several types of DRL algorithms including Deep Q Network (DQN), Deep Deterministic Policy Gradient (DDPG), and PPO. Due to the robust stability and high computational efficiency of the PPO algorithm [44], it is employed in this paper. The time sequence of lane-changing is considered, LSTM is a kind of neural network for processing sequence data and is improved based on recurrent neural networks. It solves problems of gradient disappearance and gradient explosion of recurrent neural network and can model long-term dependencies relationship of input sequence. Therefore, this paper attempts to utilize PPO and PPO-LSTM, and the final training result of PPO-LSTM proves to be superior to that of PPO, indicating that PPO-LSTM is more suitable in this paper.

PPO primarily comprises two neural networks: the actor network and the critic value network. The actor network is designed to approximate the optimal action policy, while the critic network evaluates the current state value. PPO evaluates the state after performing the action through the critic value network to update the actor network, and then uses the updated actor network to update itself. The details are shown in the green area of Fig. 2.

In this paper, the input to both networks is the data from the most recent five time steps. The critic value network outputs the estimated state value. The actor network fits the mean and standard deviation respectively, outputs the action probability distribution, and obtains the final output by sampling from the Gaussian distribution. The specific structure of the network is shown in Fig. 3. In Fig. 3(a), the standard deviation of the output is required to be limited to being greater than 0. All the activation functions in both networks use leaky ReLU.

IV. Modeling of Lane-Changing Decision-Making Model

In this section, we present the details of the Stackelberg game and the PPO-LSTM, along with the formulation of the game theory payoff function and the DRL reward function. To enhance training efficiency, this study simplifies the model by focusing solely on acceleration and deceleration behaviors instead of considering lane changing behavior of surrounding vehicles, building upon previous research [30]. Ultimately, both PPO-LSTM and Stackelberg game are employed for non-cooperative decision-making.

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A. Design of Stackelberg Game Payoff Function

In Fig. 1, when FV_2 decelerates and impacts AV's normal driving, AV must make decisions such as either reducing velocity to follow FV_2 or changing lanes for overtaking. In making these decisions, AV must take into account the reactions of surrounding vehicles RV_1 and RV_2. These reactions, which include acceleration, maintaining a constant velocity, or deceleration, directly impact subsequent decision-making by AV. For example, if AV intends to change lanes to the left, RV_1 might accelerate in order to impede this lane change. The process of this game can be modeled as a Stackelberg game. During the modeling process outlined in this paper, three key indicators are considered: driving safety, ride comfort and travel efficiency. These metrics are then aggregated with varying weights to determine the overall payoff. P^{AV} is expressed as:

$$P^{AV} = W_e^{AV}\, P_e^{AV} + W_c^{AV} P_c^{AV} + W_s^{AV} P_s^{AV} \tag{1}$$

Where P^{AV} is the total payoff of AV, P_e^{AV}, P_c^{AV} and P_s^{AV} are the travel efficiency payoff, ride comfort payoff and driving safety payoff of AV, respectively. W_e^{AV}, W_c^{AV} and W_s^{AV} are weight coefficients of travel efficiency payoff, ride comfort payoff and driving safety payoff, respectively. The importance of these three indicators can be adjusted by modifying the weight coefficients to achieve diverse driving styles [43]. The specific implementation method is described below.

The travel efficiency payoff of AV is quantified by its longitudinal velocity, which consists of two parts: the first part represents the travel efficiency influenced by FV, while the second part denotes the weighted discrepancy between AV's velocity and its maximum attainable velocity. The inclusion of this second part aims to prevent AV from maintaining the same velocity as FV and constantly tracking it. P_e^{AV} is expressed as:

$$P_e^{AV} = K_e^{AV} \begin{pmatrix} \left(1-\sigma^2\right)\left|V_x^{AV} - \tilde{V}_x^{AV}\right| \\ +\sigma^2\left|V_x^{AV} - \tilde{V}_{x\sigma}^{AV}\right| + \mu\left|V_x^{AV} - V_{\max}^{AV}\right| \end{pmatrix} \tag{2}$$

$$\tilde{V}_x^{AV} = \min\left\{V_{\max}^{AV}, V_x^{FV}\right\} \tag{3}$$

$$\tilde{V}_{x\sigma}^{AV} = \min\left\{V_{\max}^{AV}, V_{x\sigma}^{FV}\right\} \tag{4}$$

Where σ represents the lane-changing behavior, $\sigma \in \{-1, 0, 1\}$ = {change left, no change, change right}. V_x^{AV} denotes the longitudinal velocity of AV after time *t*. $V_{\max}^{AV}$ is the maximum velocity of AV. V_x^{FV} is the longitudinal velocity of FV in the lane where AV is located. $V_{x\sigma}^{FV}$ is the longitudinal velocity of FV in AV's adjacent lane. μ is a discount factor that adjusts expectations for maximum velocity. K_e^{AV} adjusts the magnitude range of the driving efficiency payoff to ensure that the three indicators are on the same scale. The equation representing the K_e^{AV} is given by:

$$K_e^{AV} = \frac{1}{(\mu+1)\left(V_{\max} - V_{\min}\right)} \tag{5}$$

Where $V_{\max}$ and $V_{\min}$ represent the maximum and minimum velocity limit. The ride comfort payoff of AV can be quantified by analyzing the acceleration parts, namely longitudinal and lateral accelerations. P_c^{AV} is expressed as:

$$P_c^{AV} = \frac{\max\limits_{1\le i\le n}\left\{\sqrt{\left(a_{x-i}^{AV}\right)^2 + \sigma^2\beta\left(a_{y-i}^{AV}\right)^2}\right\}}{a_{\max}^{AV}} \tag{6}$$

Where a_{x-i}^{AV} denotes the longitudinal acceleration of AV, a_{y-i}^{AV} represents its lateral acceleration, and β is a weighting coefficient for lateral acceleration. The maximum acceleration of AV is denoted as $a_{\max}^{AV}$, and $\max_{1\le i\le n}$ indicates the solution for achieving maximum acceleration within the predicted time interval *t*. The driving safety payoff of AV is determined by both relative velocity and relative distance, with a smaller payoff indicating a higher level of safety. When making lane changes, AV takes into account FV and RV in AV's adjacent lane while engaging in a game with RV. P_s^{AV} is expressed as:

$$\begin{aligned} P_s^{AV} = {} & \left(1-\sigma^2\right) S_{AV}^{FV} \\ & + \sigma^2\left(S_{AV}^{\sigma FV} + S_{\sigma FV}^{AV}\right) + \frac{\mathrm{s}}{X_x^{FV} - X_x^{AV} - L_{AV}} \end{aligned} \tag{7}$$

$$S_{AV}^{FV} = \frac{\left|T_0 - \min\left\{\left|\frac{X_x^{FV} - X_x^{AV} - L_{AV}}{V_x^{AV} - V_x^{FV} + \varsigma}\right|, T_0\right\}\right|}{T_0} \tag{8}$$

Where S_{AV}^{FV} represents the driving safety payoff between AV and FV in AV's current lane after time *t*, $S_{AV}^{\sigma FV}$ and $S_{\sigma RV}^{AV}$ represent the driving safety payoff between AV and FV in AV's adjacent lane, as well as between AV and RV in AV's adjacent lane, respectively, after time *t*. The calculation method of $S_{AV}^{\sigma FV}$ and $S_{\sigma RV}^{AV}$ is the same as that of S_{AV}^{FV}. X_x^{AV}, X_x^{FV}, $X_{x\sigma}^{FV}$ and $X_{x\sigma}^{RV}$ denote respectively the longitudinal positions of AV, FV in AV's current lane, FV in an adjacent lane to AV, and RV in an adjacent lane to AV after time *t*. $V_{x\sigma}^{RV}$ represents longitudinal velocity of RV in an adjacent lane to AV while L_{AV} denotes length of AV. T_0 is a threshold value for safe time distance between two vehicles valued at 3 [45]. A value greater than T_0 indicates a safe condition.

The calculation method for RV's payoff is identical to that of AV payoff P^{AV}. The coefficients for the three indicators—travel efficiency, ride comfort, and driving safety—are all set to 1 in this paper. Although the coefficients of the three indicators were set to 1 in this paper, the agent was still able to perceive the surrounding environment information and consider the personalized information of RV when calculating the AV payoff, thus making the optimal decision-making.

B. Solution of the Game

In this section, we aim to determine the optimal strategy in a Stackelberg game by the aforementioned constructed model, where AV assumes the role of leader and RV acts as the follower. This solution can be regarded as a two-layer nonlinear optimization problem. However, considering computational efficiency [46], [47], in this paper, the accelerations of the AV and the SVs in the solution space are uniformly discretized with an interval of 0.5 (where $a_x^{HV} \in \{-3\ \mathrm{m/s^2}, -2.5\ \mathrm{m/s^2}, -2\ \mathrm{m/s^2}, -1.5\ \mathrm{m/s^2}, -1\ \mathrm{m/s^2}, -0.5\ \mathrm{m/s^2}, 0\ \mathrm{m/s^2}, 0.5\ \mathrm{m/s^2}, 1\ \mathrm{m/s^2}, 1.5\ \mathrm{m/s^2}, 2\ \mathrm{m/s^2},$

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2.5 m/s^2,3 m/s^2}). Thus, the two-layer nonlinear optimization problem is transformed into a discrete optimization problem and can be solved using grid search. Our model is able to play games with the two surrounding RVs (in the left and right lanes) and can also automatically determine whether to engage in a game with one or both of them based on the situation of the surrounding vehicles. Such a setting is relatively in line with the real situation [30]. The forms of the game in this paper are categorized into two types: the first type is when only one side can change lanes, and the second type is when both sides can change lanes.

In the first type, AV engages in a two-player game with a rear vehicle in an adjacent lane. The solution for the first type is as follows:

$$\left(a_x^{HV*},\sigma^*\right)=\arg\min_{a_x^{AV},\sigma}\left(\max_{a_{x\sigma}^{RV}\in\gamma^2\left(a_x^{AV},\sigma\right)} P^{AV}\left(a_x^{AV},\sigma,a_{x\sigma}^{RV}\right)\right) \tag{9}$$

$$\gamma^2\left(a_x^{AV},\sigma\right)= \left\{\xi\in\Gamma^2 : P_\sigma^{RV}\left(a_x^{AV},\sigma,\xi\right)\le P_\sigma^{RV}\left(a_x^{AV},\sigma,a_{x\sigma}^{RV}\right),\forall a_{x\sigma}^{RV}\in\Gamma^2\right\} \tag{10}$$

subject to

$$\begin{cases}\sigma\left(\sigma-1\right)\left(\sigma+1\right)=0\\ V_{\min}\le V_x^{AV}\le V_{\max}\\ V_{\min}\le V_{x\sigma}^{RV}\le V_{\max}\end{cases} \tag{11}$$

Where P_σ^{RV} represents total payoff of RV in adjacent lanes of AV, $a_{x\sigma}^{RV}$ represents the acceleration of RV in adjacent lanes of AV, and $V_{x\sigma}^{RV}$ represents the velocity of the RV in adjacent lanes of AV. $V_{\max}$ stands for maximum velocity limit while $V_{\min}$ represents minimum velocity limit. $\gamma^2(a_x^{AV},\sigma)$ represents the optimal policy candidate set for RV given an AV policy. Γ^2 represents the set of all possible policies for RV.

The second type is when both the left and right lanes of AV are present, as shown in Fig. 1. AV needs to consider RV$_1$ and RV$_2$. Due to the significantly higher computational complexity of multiplayer games compared to two-player games, and the relatively small interaction between RV$_1$ and RV$_2$ in practical scenarios, we treat them as two separate two-player game problems. The final result is to choose the smaller one among the gains obtained from solving two game theories. Equation (9) should be rewritten as:

$$\left(a_x^{AV*},\sigma^*\right)=\arg\min_{\left(a_x^{AV*},\sigma^*\right)\in U} \left(P^{AV}\left(a_{xl}^{AV*},\sigma^{l*},a_{x\sigma}^{RV1}\right),P^{AV}\left(a_{xr}^{AV*},\sigma^{r*},a_{x\sigma}^{RV2}\right)\right) \tag{12}$$

$$U\in\left\{\left(a_{xl}^{AV*},\sigma^{l*}\right),\left(a_{xr}^{AV*},\sigma^{r*}\right)\right\} \tag{13}$$

The meanings of constraints and parameters are the same as those in (9), so we do not elaborate them here.

C. State and Action

The state information in DRL reflects the changes induced by environmental variables perceived by the agent and the actions generated by the agent itself. This state information serves as a fundamental basis for agents to evaluate long-term decisions and make optimal choices. In this paper, the state variable primarily encompasses the data that an autonomous vehicle perceives and processes from the environment. The processed data is then taken as the state variable. There are primarily two types of data: relative velocity and relative distance, considering both AV in the current lane and the FV and RV in the adjacent lane. All the processed environmental variables are combined to form the state variable set S.

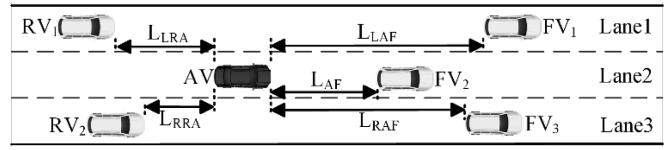

Fig. 4. Definition of variables for the agent.

$$S=\{L_{LRA},L_{RRA},L_{AF},L_{LAF},L_{RAF},V_{LRA},V_{RRA},V_{AF}, V_{LAF},V_{RAF}\} \tag{14}$$

Where L_{LRH} represents the longitudinal distance between the center of the rear axles of RV$_1$ and AV; L_{LAF} represents the longitudinal distance between the center of the rear axles of AV and FV$_1$; similarly, L_{RRA}, L_{AF} and L_{RAF} have analogous interpretations. These details are shown in Fig. 4. V_{LRA} denotes the relative velocity between RV$_1$ and AV, and similarly, V_{RRA}, V_{AF}, V_{LAF}, and V_{RAF} represent the relative velocities between their respective vehicles.

In this paper, we propose combining PPO-LSTM with the Stackelberg game to dynamically adjust the weights assigned to travel efficiency payoff, ride comfort payoff and driving safety payoff. Consequently, we define the action space of PPO-LSTM as continuous. The actor network outputs both the mean and standard deviation of a Gaussian distribution and performs non-boundary sampling from this distribution. The action space may be defined as:

$$a=\left\{W_e^{AV},W_c^{AV},W_s^{AV}\right\} \tag{15}$$

It should be noted that all the output actions fall within the range of (0, 1), and they have the following relationship:

$$W_e^{AV}+W_c^{AV}+W_s^{AV}=1 \tag{16}$$

D. Reward Function

In this paper, the design of the reward function is improved upon the payoff function from game theory, enabling the agent to simultaneously consider travel efficiency, ride comfort, and driving safety during the learning process. Besides, it complements the design of the payoff function in game theory. For the agent to stably learn an optimal strategy, the reward is typically a positive value [48]. Therefore, in this paper, we appropriately transform the three key indicators, such as P_e^{AV}, P_c^{AV} and P_s^{AV}, from the Stackelberg game returns. We employ larger values to create a buffer that guarantees r remains positive as depicted in (19). Additionally, to avoid AV from colliding with other vehicles due to inappropriate proportions between various indicators, a

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collision avoidance loss is introduced to avoid the agent from behaving inappropriately.

$$r = 5 - \left(P_e^{AV} + P_c^{AV} + P_s^{AV}\right)^2 + P_{exist}^{AV} \tag{17}$$

Where P_{risk}^{AV} aims to avoid dangerous driving behaviors caused by collisions between AV and surrounding vehicles. P_{risk}^{AV} is expressed as:

$$P_{exist}^{AV} = \begin{cases} -100, & collision \\ 0, & else \end{cases} \tag{18}$$

In the process of reward function design, P_c^{AV} can be used not only to maximize driving comfort, but also to avoid frequent lane changes through lateral ride comfort.

V. Training and Simulation Testing

This section begins by detailing the simulation environment, followed by a presentation of the relevant parameters, evaluation metrics, and training outcomes. Lastly, the results are tested and analyzed across various scenarios.

A. Simulation Settings

Considering safety, efficiency, and other factors, this study utilized Matlab, Simulink, and Simulation of Urban MObility (SUMO) [49] for simulation and training. To enhance the operational efficiency of the model, the vehicle dynamics model was simplified to a two-wheel bicycle model [32]. A fifth-order polynomial was employed for path planning, while the Stanley method was used to control the vehicle steering angle for achieving precise path tracking. To evaluate the model performance more comprehensively, three cases were designed: two preset scenarios to test the model's limit performance and approximately 100 real-world-like scenarios to assess its generalization and stability. The settings of these three cases are as follows.

Case 1: The scenario in Case 1 involves five SVs. Fig. 5(a) illustrates the initial state of this scenario, where the AV is surrounded by the SVs. The initial velocity of the AV is 24 m/s. Fig. 5(b) presents the velocity curves of all the SVs, with the simulation duration being 10 seconds. This case was evaluated using Simulink.

Case 2: The AV travels at an initial velocity of 24 m/s. Six SVs are positioned ahead, all moving at a constant velocity of 21 m/s. The AV sequentially overtakes the six SVs. Fig. 5(c) illustrates the initial state of this scenario. The simulation duration for this case is 20 seconds and is conducted using Simulink.

Case 3: While both Case 1 and Case 2 examine the model's performance under challenging conditions, they overlook the more prevalent real-world scenarios that are typically encountered. Therefore, in this paper, drawing on the study [50], the proportions of aggressive, normal, and conservative driving styles were set in SUMO to reflect real-world conditions (13%, 41%, and 46%), respectively. Subsequently, based on the paper [28], these three driving styles were implemented using Intelligent Driver Model (IDM) parameters and lane-changing cooperation coefficients. The road consists of three lanes, with a vehicle generation rate of 1 vehicle per second (equivalent to a traffic flow of 3600 veh/h). Lateral control for surrounding vehicles is executed using the built-in LC2013 model in SUMO. Prior to each testing round, the simulation scene requires a 20-second preheating period to stabilize environments. This case involves a co-simulation test conducted jointly by Matlab, Simulink, and SUMO.

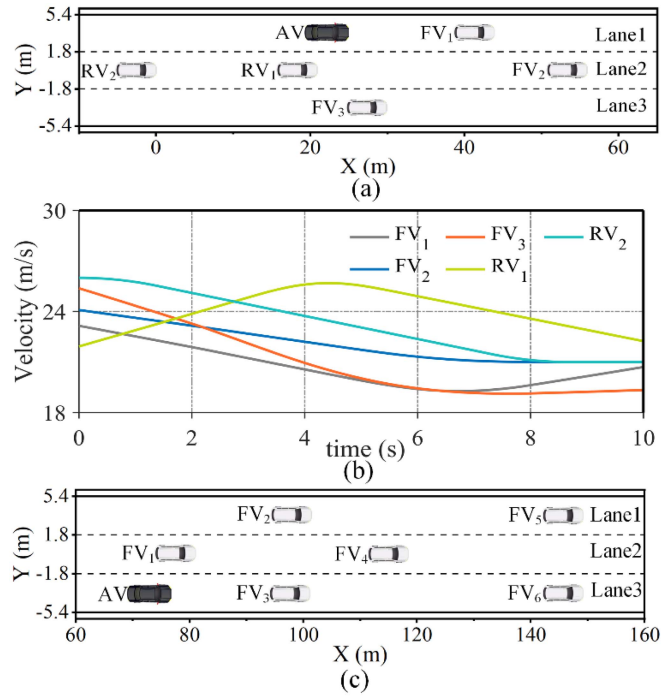

Fig. 5. Initial states and velocities of the test scenarios. (a) Shows the initial state of the scenario in Case 1, (b) shows the velocity of the vehicle in Case 1, and (c) shows the initial state of the scenario in Case 2.

TABLE II
Parameters of DRL and Simulation

Components	Parameters	Values
DRL	Discount factor	0.96
	Learning rate of Actor network	0.0007
	Learning rate of Critic network	0.0008
	Clip factor	0.2
	EntropyLossWeight	0.01
	MiniBatchSize	128
	ActorOptimizerOptions	Adam
	CriticOptimizerOptions	Adam
Simulation	Max step	350
	SampleTime	0.1s

B. Model Training

In this paper, the parameters of simulation and DRL are shown in Table II, and the training scenarios are randomly combined from Case 1 and Case 2. We conducted three training sessions with distinct random seeds to minimize the impact of randomness. The resulting training outcomes are presented in Fig. 6. The dotted and solid lines in the Fig. 6 represent the average rewards of the three training results, while the light-colored areas denote the 95% confidence intervals. Overall,

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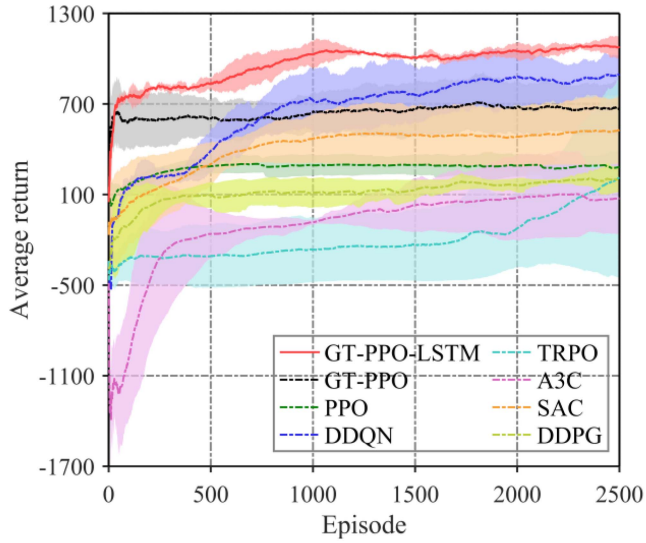

Fig. 6. Training results.

the GT-PPO-LSTM proposed in this paper converges effectively to an optimal level. Furthermore, since the decision-making main parts of GT-PPO-LSTM and GT-PPO are based on game theory, their decision-making before training is superior to that randomly generated initially in DRL, resulting in higher rewards for the initialization decision-making. The superior training outcomes of GT-PPO-LSTM compared to GT-PPO suggest that incorporating LSTM to account for lane-changing timing enhances decision-making quality. The rapid convergence of rewards during training underscores the rationality of the GT-PPO-LSTM-based model and validates the effectiveness of the reward function design presented in this paper.

C. Model Comparison

This section primarily describes the models compared against the GT-PPO-LSTM proposed in this study. The primary objective is to validate the effectiveness of our model and assess the contribution of each module within the model. In this paper, the proposed model is compared with GT-PPO, aggressive GT, conservative GT, and normal GT to validate the effectiveness of the PPO and LSTM modules in GT-PPO-LSTM. The payoff calculation formulas for GT are identical to those in GT-PPO-LSTM, and distinct driving styles are implemented as outlined in Table I. Subsequently, the model's performance was validated by comparing it with several state-of-the-art algorithms in the field. In this paper, MOBIL and IDM (hereinafter collectively referred to as rule-based models for simplicity), DDQN [51], A3C [52], PPO [44], SAC [53], TRPO [54], and DDPG [55] are selected as benchmark models. Specifically, MOBIL is a classic lane-changing decision-making model, while IDM is a mature longitudinal control model. The rule-based models integrate these two models. DDQN can address the problem of overestimating Q values, thus achieving better performance. A3C, as a variant of the classic policy gradient algorithm Actor-Critic (AC), has high computational efficiency. PPO is widely used across various fields due to its stability. SAC introduces maximum entropy regularization to balance exploration of the unknown with exploitation of known optimal actions. TRPO adopts trust region constraints to control the step size of policy updates and ensure the reliability of these updates. These algorithms are widely recognized as benchmark models. Similar to the model proposed in this paper, they make collaborative decisions both laterally and longitudinally and control velocity through acceleration adjustments. It is worth noting that the acceleration settings are consistent with those of GT-PPO-LSTM. In addition, DDPG has a high convergence speed and is good at dealing with continuous action spaces. We use DDPG for continuous velocity control and assign lateral control to MOBIL (Use IDM to calculate the acceleration benefit). The purpose is to verify that our discretization of acceleration is acceptable.

D. Evaluation Metrics

This paper primarily evaluates efficiency and safety. Key indicators considered include velocity' as a measure of efficiency, time headway (THW) and time to collision (TTC) as measures of safety, as well as the success rate during testing, defined by the frequency of no collisions. The THW refers primarily to the average value measured during the testing process. A threshold of 3 seconds is established for TTC, below which potential risks are deemed to exist [45]. Both TTC and THW can more comprehensively indicate the safety level between AV and FV. In this paper, all models were tested under the same time conditions. Driving distance refers to the distance that the vehicle travels safely. The longer the driving distance, the higher the vehicle efficiency, and it can also indirectly reflect the safety performance of the model. Furthermore, to more vividly demonstrate the driving style adaptive adjustment capability of the GT-PPO-LSTM proposed in this paper, we quantified the driving style. Given that the HighD [56] dataset is a highway dataset and aligns closely with the research context of this paper, we utilized the HighD dataset for driving style clustering. Extract the data 2 seconds prior to the lane-changing moment for vehicles performing a lane change [57], where the FV exists and is located no more than 100 meters ahead. During training, the average value of these extracted data points is utilized [27]. This setup primarily aims to align with the test cases employed in this paper. The features utilized in this paper include longitudinal and lateral relative distances, relative velocities, and relative accelerations with respect to the FV [40]. These six features effectively capture the urgency of lane-changing maneuvers as well as the following distance, among other factors. Since the clustering of driving styles is not the main research focus of this paper, the method chosen is K-Means, which is widely recognize [40]. The clustering results are shown in Fig. 7, showing two main indicators: relative velocity and relative distance. This paper employs this model to quantify the driving style of AVs during the testing process. Given that driving style reflects behavior over a period of time, this paper adopts a time-window-based approach for driving style recognition to avoid frequent fluctuations in classification [58]. Specifically, the time window spans 1 second before and 1 second after the

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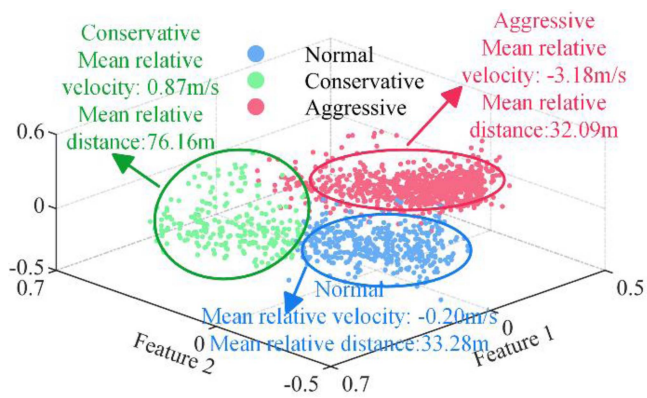

Fig. 7. Results of driving style recognition. The relative value is FV minus AV.

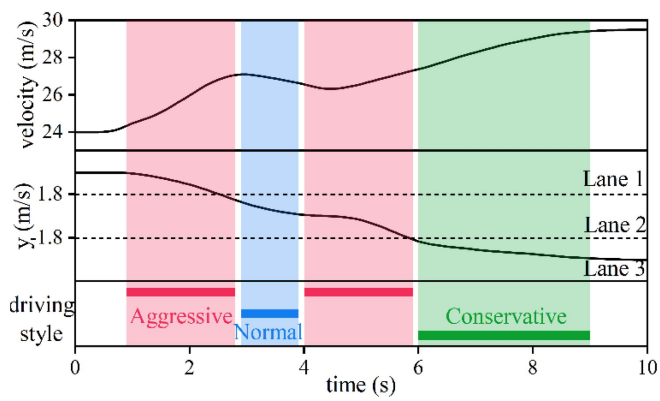

Fig. 8. The detailed results in Case 1.

current moment. By applying a sliding time window, a sequence of driving styles can be systematically obtained.

E. Model Evaluation

In this section, we evaluated all the models under three cases and conducted a detailed analysis of the results. Specifically, among the models selected for this paper, some require training while others do not. Given that the training process involves stochasticity, for the models requiring training, we present both the average of the three trials and the standard deviation. In contrast, for the models that do not require training, we directly report the test results.

1) Case Study 1: This paper evaluated all the models under the Case 1, where the primary objective was to assess the limit performance of the models in complex and challenging scenarios. The Table III shows the test results of all models. It is worth noting that, given the deterministic nature of the individual game theory model and the rule-based model, which do not involve training or randomness. Consequently, there is no need to present these results using mean and standard deviation. During the testing process, the GT-PPO-LSTM model proposed in this paper demonstrated superior performance by successfully navigating the scenario with the highest efficiency and a relatively high level of safety. It achieved an average velocity of 27.02 m/s, with its THW ranking second only to the rule-based model at 0.91 seconds. Notably, throughout the testing process of Case 1, the TTC remained consistently above the threshold, showcasing safety levels comparable to those of the safest rule-based model. Rule-based, TRPO, A3C, Normal-GT, DDPG, and Conservative-GT all adopted relatively conservative deceleration strategies and followed FV_1. Among these models, only the Normal-GT and TRPO models performed a lane change at the end of the simulation, entering between RV_1 and RV_2 and subsequently following RV_1.

Consequently, this explains why Normal-GT and TRPO exhibit the highest THW among the models that implement conservative decision-making strategies. While adopting a conservative strategy enhances safety, it also significantly reduces efficiency. The GT-PPO-LSTM model proposed in this paper is different from the conservative strategy. Specifically, at the initial stage, it executed an accelerated lane change to overtake RV_1, thereby preventing itself from being trapped between FV_1, RV_1, and RV_2. This is because, as illustrated in Fig. 5, all three vehicles will decelerate, which forces the conservative strategy to also decelerate and delays its entry into Lane 2. Subsequently, the AV will decelerate to identify a suitable opportunity for continuing its lane change to Lane 3, thus efficiently navigating through this scenario. While the driving methods of Aggressive-GT, GT-PPO, and DDQN resemble those of GT-PPO-LSTM, they fail to match the safety and efficiency levels achieved by GT-PPO-LSTM.

Regarding the effectiveness of each module in GT-PPO-LSTM, the test results demonstrate that GT-PPO-LSTM outperforms GT-PPO, thereby validating the effectiveness of incorporating LSTM as proposed in this paper. Additionally, GT-PPO-LSTM surpassing the three driving style models implemented by GT further highlights the benefits of incorporating PPO-LSTM within the framework of game theory. Table III also illustrates that GT outperforms certain RL models.

Regarding the driving style of AV, Fig. 8 illustrates the velocity and trajectory during the test. As shown in the Fig. 8, at the outset, the agent prioritizes efficiency by increasing its weight, leading AV to adopt a more aggressive driving style for acceleration and lane-changing to overtake RV_1. Subsequently, due to the increased complexity of the environment ahead of FV_2 and RV_1, the agent adjusts its priorities by reducing the weight of efficiency and enhancing the weight of safety, thus shifting the driving style toward a normal style. During this phase, the velocity decreases as AV seeks opportunities for lane changes and overtaking. Finally, after all vehicles have been overtaken, the agent increases the weights of comfort and safety, resulting in AV adopting a more conservative driving style or providing a smoother driving experience.

Overall, the GT-PPO-LSTM model proposed in this paper demonstrates the highest efficiency and achieves safety performance that ranks second only to the rule-based model in Case 1.

2) Case Study 2: In Case 2, SVs travel at a uniform velocity with consistent gaps between them. This scenario primarily requires the model to accurately perceive the environmental conditions of the current scene and subsequently generate appropriate decisions. Table IV presents the test results for all models

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TABLE III
MODEL EVALUATION RESULTS IN CASE 1

Model	Average THW(s)	TTC<3	Driving distance(m)	Average velocity(m/s)
Aggressive-GT	0.8183	4.95%	**269.13**	**26.91**
Conservative-GT	0.7069	**0**☆	207.56	20.76
Normal-GT	0.6508	**0**☆	220.09	22.01
GT-PPO	0.7883±0.04	3.63%±1.14	262.91±5.62	26.29±0.56
Rule-based	**0.9203**☆	**0**☆	207.76	20.78
DDPG	0.85±0.04	0±0	211.53±1.49	21.15±0.15
A3C	0.91±0.03	**0±0**☆	209.66±2.91	20.97±0.29
DDQN	0.87±0.07	0.66%±0.57%	268.55±2.54	26.86±0.25
PPO	0.86±0.02	**0±0**☆	244.55±5.43	24.46±0.54
SAC	0.87±0.15	**0.33%±0.57%**	246.55±14.55	24.66±1.46
TRPO	0.67±0.23	1.32%±1.51%	221.78±17.62	22.18±1.76
GT-PPO-LSTM	**0.91±0.02**	**0±0**☆	**270.21 ±2.65**☆	**27.02±0.27**☆

The ones with asterisks are the best results, and only the underlined and bold ones are the second-best results.

TABLE IV
MODEL EVALUATION RESULTS IN CASE 2

Model	Average THW(s)	TTC<3	Driving distance(m)	Average velocity(m/s)
Aggressive-GT	0.8576	3.84%	**493.37**	**24.67**
Conservative-GT	0.6827	**0***	432.43	21.62
Normal-GT	0.7239	**1%**	449.01	22.45
GT-PPO	0.88±0.03	1.16%±1.15%	480.22±21.74	24.01±1.09
Rule-based	**1.42**☆	**0**☆	408.19	20.41
DDPG	**1.18±0.03**	0±0	417.26±3.43	20.86±0.17
A3C	1.02±0.42	2.99%±1.32%	450.51±16.61	22.53±0.83
DDQN	1.02±0.09	**0±0**☆	420.99±7.55	21.05±0.38
PPO	0.83±0.17	9.78%±1.04%	482.13±11.65	24.11±0.58
SAC	1.04±0.04	**0±0**☆	419.72±3.05	20.99±0.15
TRPO	0.90±0.13	3.98%±0.68%	469.32±15.61	23.46±0.78
GT-PPO-LSTM	1.05±0.01	1.66%±0.29%	**498.95±3.66**☆	**24.95±0.18**☆

The ones with asterisks are the best results, and only the underlined and bold ones are the second-best results.

in Case 2. According to the results shown in Table IV, it is evident that the GT-PPO-LSTM proposed in this paper achieves the highest efficiency while maintaining a relatively high level of safety. During testing, the average velocity was 24.95 m/s, the driving distance was 498.95 m, THW was 1.05 seconds, and the proportion of TTC values greater than 3 seconds was lower compared to Aggressive-GT, which exhibited similar efficiency. The AVs utilizing Conservative-GT, Rule-based, SAC, DDPG, and DDQN continuously decelerated to maintain a safe distance from the preceding vehicle FV_3. These models adopted highly conservative driving strategies and refrained from executing lane changes throughout the Case 2 test. In contrast, Normal-GT, A3C, and TRPO employed lane-changing maneuvers to enhance efficiency. However, they were unable to fully succeed in completing the scenario. The AV using A3C and TRPO eventually positioned itself between FV_2 and FV_5 during the test period, whereas the AV using Normal-GT remained behind FV_4. Although these models exhibit higher efficiency compared to the aforementioned conservative models, this improvement comes at the cost of reduced safety. Specifically, the proportion of cases in which TTC is less than 3 seconds has increased significantly. The GT-PPO-LSTM proposed in this study demonstrates strong performance in handling the described scenario. It effectively observes the surrounding environment to determine optimal lane-changing timing, such as accelerating while overtaking FV_3 and subsequently decelerating to follow FV_4. Ultimately, it achieved a result that was second only to the rule-based model THW, with a score of 1.05 seconds. Moreover, the proportion of TTC values greater than 3 seconds was lower compared to all models capable of successfully completing this scenario. In contrast, although Aggressive-GT, GT-PPO, and PPO could also complete the scenario, they exhibited lower THW scores and a higher proportion of TTC values reaching dangerous levels.

Regarding the effectiveness of each module in GT-PPO-LSTM, a comparison of the models for the three driving styles implemented by GT and the results of GT-PPO demonstrates that integrating PPO and LSTM into GT yields significant improvements. By comparing GT-PPO with GT-PPO-LSTM, we find that GT-PPO-LSTM exhibits higher security and efficiency, which indicates that the LSTM module is effective. Moreover, GT's performance in this context outperforms several RL models.

Regarding the driving style of the AV, Fig. 9 illustrates the velocity and trajectory during the testing process. At the

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TABLE V
TEST RESULTS UNDER SCENARIOS INVOLVING VARIOUS DRIVING STYLES

Model	Success rate	Average velocity(m/s)	Average THW(s)
Aggressive-GT	95%	**28.76**	1.49
Conservative-GT	98%	27.54	1.61
Normal-GT	98%	28.03	1.57
GT-PPO	90.67%±2.31%	28.06±0.16	1.44±0.07
Rule-based	**100**☆	26.78	**2.4069**
DDPG	93.67±2.31%	26.61±0.23	1.94±0.20
A3C	89.67%±1.15%	20.99±0.26	**2.78±0.24**☆
DDQN	27.41±0.80	27.41±0.80	1.39±0.08
PPO	36.00%±2%	21.71±0.78	1.59±0.17
SAC	86.33%±2.52%	22.09±0.41	2.35±0.13
TRPO	75.33%±8.32%	24.13±1.13	1.82±0.31
GT-PPO-LSTM	**99.67%±0.57%**	**28.94±0.34**☆	1.68±0.05

The ones with asterisks are the best results, and only the underlined and bold ones are the second-best results.

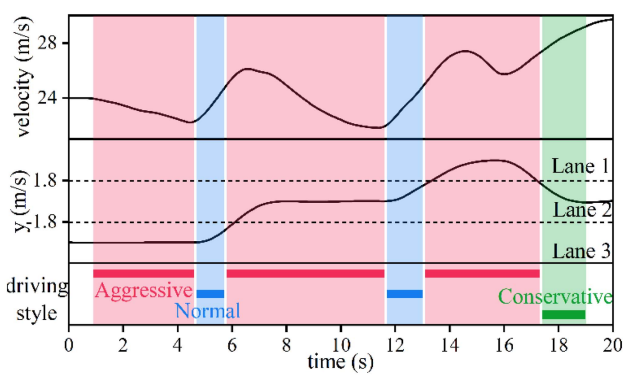

Fig. 9. The detailed results in Case 2.

outset, the AV gradually decelerates to interact with the FV_1 and determine the necessity of a lane change. During this phase, as the AV continuously approaches the FV_3, the TTC and THW progressively decrease. Consequently, the agent adjusts its decision-making by increasing the weight on efficiency, thereby shifting the driving style toward a more aggressive style to successfully execute the maneuver. Subsequently, the agent prioritizes safety by increasing its weight in decision-making. Given that FV_3 and FV_1 are traveling at the same velocity, the agent executes a lane change while accelerating to create distance from FV_1, thereby ensuring it is not rear-ended during the maneuver. Following this, it decelerates gradually to close in on FV_4 and search for an opportunity to change lanes. After entering Lane 1 and overtaking FV_4, the agent decelerates once more to avoid rear-ending FV_5 before transitioning into Lane 2.

Overall, the GT-PPO-LSTM not only achieves the highest driving efficiency but also maintains a relatively high level of safety.

3) Case Study 3: For Case study 3, the primary objective is to evaluate the model's generalization capability and its proficiency in handling conventional scenarios. The scenarios in Case 3 were randomly generated using SUMO and were not encountered during the training phase. Furthermore, this case involves SVs exhibiting diverse driving styles, which allows for an assessment of the model's decision-making ability when confronted with multiple SVs displaying varied driving behaviors simultaneously. Table V presents the test results under scenarios involving various driving styles. The TTC values for all models in this case of scenario generally remained below the safety threshold. In other words, from a TTC perspective, all models can be considered relatively safe. Therefore, further analysis of TTC was not pursued. It can be observed from Table V that the GT-PPO-LSTM demonstrates the highest efficiency among all models. Although its THW ranks sixth in the test results at 1.68 seconds, the success rate of GT-PPO-LSTM across 100 scenarios achieves the second-highest level, averaging 99.67%, with two instances reaching 100%. This further suggests that the safety performance of GT-PPO-LSTM is satisfactory. The rule-based model achieves a higher THW because longitudinal control is governed by the IDM. This model prioritizes parameters such as expected THW and expected distance, resulting in the maintenance of a larger THW while matching the safety velocity. However, test results indicate that the average velocity of the rule-based model is only 26.78 m/s, which is lower than that of the GT-PPO-LSTM model. DDPG can achieve a relatively high success rate, but this comes at the expense of efficiency. However, in terms of both the success rate and average speed, it is still lower than GT-PPO-LSTM, indicating that the discretization of acceleration in this study is acceptable. For A3C, SAC, PPO, their generalization ability is limited, and they tend to decelerate significantly under challenging conditions, often maintaining speeds close to the minimum limit. Other models enhance efficiency by adopting strategies such as lane changes; however, this comes at the cost of reduced safety. While the efficiency of these other models remains relatively high, their performance in ensuring safety is notably weaker. Moreover, GT-PPO-LSTM demonstrates robust adaptability in Case 3, where three driving styles are mixed. In contrast, the benefit function of SV within the GT-implemented models for the three driving styles lacks the flexibility to accommodate diverse driving scenarios, thereby compromising safety.

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Furthermore, after incorporating LSTM into GT-PPO, although the model's efficiency did not exhibit a substantial improvement, there was a significant enhancement in safety. This suggests that integrating LSTM into GT-PPO is effective. When compared with the game theory models of the three driving styles, GT-PPO-LSTM demonstrates improvements in both efficiency and safety, indicating that the integration of PPO and LSTM within a game theory framework is indeed effective. In Case 3, given that 100 test rounds were conducted and the driving style could not be consistently analyzed in alignment with the previous cases, the analysis of the driving style was omitted for this case.

Overall, the GT-PPO-LSTM model proposed in this study demonstrates excellent efficiency and safety in complex scenarios, exhibits strong generalization capabilities in untrained scenarios during the training process, and thereby achieves ideal performance.

The analysis of three cases shows that the GT-PPO-LSTM model proposed in this paper maintains high efficiency while ensuring a high level of security, and it outperforms the baseline model. Regarding the discrete processing of acceleration, although DDPG can achieve a relatively high level of safety, it has extremely low efficiency. GT-PPO-LSTM outperforms DDPG in terms of both overall safety and efficiency, indicating that this study can accept the degree of discretization of acceleration. In addition, according to the ablation experiments, all modules (GT, PPO, LSTM) in the model proposed in this study contribute to its performance. The comparison among GT-PPO-LSTM, GT-PPO, GT, and PPO illustrates the effectiveness of the two approaches—DRL and game theory—and confirms that more efficient decision-making can be achieved through interactive modeling.

VI. Conclusion

Regarding the issues of game theory's inability to adapt to complex and dynamic environments, this paper proposes a lane-changing decision-making model (GT-PPO-LSTM) that combines Stackelberg game with PPO-LSTM interactive modeling, aiming to achieve clear causal relationships in the decision-making process and adaptability to complex and dynamic environments. By leveraging the agent's perception ability of the environment to dynamically adjust the importance of three major indicators: driving safety, ride comfort, and travel efficiency within the game framework, the agent intervenes in the calculation process of the game payoff function to control fitting different driving styles for AVs and make decisions that are most suitable for the current environment. The test results show that GT-PPO-LSTM can achieve excellent performance in terms of efficiency and safety in different scenarios, and can adjust the driving style of AV according to different environments for more efficient decision-making. In addition, in the case where the main body of the decision-making framework is game theory, it can maintain a clear causal relationship in the decision-making process. The proposed interactive modeling approach in this paper effectively integrates the advantages of both Stackelberg game and PPO-LSTM, addressing issues such as the difficulty of adapting to the dynamic environment in game theory and the unclear decision-making process in DRL.

The method proposed in this paper incorporates the advantages of game theory. However, it does not address its shortcomings, such as slow solving speed, and requires enhancement for more complex scenarios. In future work, it is anticipated that this model will be further improved to enhance solving speed and expand its applicability to more intricate and diverse scenarios.

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